SPCMP DETERMINISM REPORT

Samayuktam Profile for PCMP v1 | Generated 2026-05-19 06:09 UTC

Executive Summary

Test conducted across 2 hardware target(s): CPU, GPU. Three tests on an identical 1M-element float32 vector. top_k and bottom_k test SPCMP ordering convergence (algorithm-driven ordering divergence). scalar_reduction documents the protocol boundary (value-level FP divergence).

Test	Goal	Raw unique	SPCMP unique	Verdict
Top-K Selection	SPCMP converge	2	1	PASS
Scalar Reduction	SPCMP diverge (limit)	2	2	PASS
Bottom-K Selection	SPCMP converge	2	1	PASS

Hardware Fingerprints

CPU — 20260519T060630Z

Field	Value
compute_used	CPU -- numpy (GPU bypassed by --hw cpu)
platform	Linux-6.6.122+-x86_64-with-glibc2.39
cpu	Intel(R) Xeon(R) CPU @ 2.20GHz
torch	None
cuda_device	None
jax_version	None
jax_devices	none

GPU — 20260519T060652Z

Field	Value
compute_used	CUDA -- NVIDIA GeForce RTX 3050 6GB Laptop GPU
platform	Linux-6.6.87.2-microsoft-standard-WSL2-x86_64-with-glibc2.39
cpu	12th Gen Intel(R) Core(TM) i5-12450HX
torch	2.12.0+cu130
cuda_device	NVIDIA GeForce RTX 3050 6GB Laptop GPU
jax_version	None

jax_devices	none

SHA-256 Cross-Hardware Comparison

Top-K Selection

PASS — raw ordering diverged, SPCMP converged

Hardware	Device / Algo	Raw SHA-256	=?	SPCMP SHA-256	=?
CPU (CPU -- numpy (GPU bypassed by --hw c…) CPU: Intel(R) Xeon(R) CPU @ 2.20GHz	cpu (numpy) np.argpartition (heap order)	f76e470f5b79…1ab2d5c4	=	bf476da80bcb…a9f39551	=
GPU (CUDA -- NVIDIA GeForce RTX 3050 6GB …) CPU: 12th Gen Intel(R) Core(TM) i5-12450HX	cuda:0 (NVIDIA GeForce RTX 3050 6GB Laptop GPU) torch.topk(largest=True, sorted=True) -> descending abs order	3ad9c18dbd00…c666ae1a	≠	bf476da80bcb…a9f39551	=

Scalar Reduction

PASS — value divergence confirmed; SPCMP correctly unable to resolve

Note: SPCMP divergence is the CORRECT outcome here — it proves the protocol is honest about its limits.

Hardware	Device / Algo	Raw SHA-256	=?	SPCMP SHA-256	=?
CPU (CPU -- numpy (GPU bypassed by --hw c…) CPU: Intel(R) Xeon(R) CPU @ 2.20GHz	cpu (numpy)	3714dbf09947…c336978f	=	66439ad3daef…66866665d	=
GPU (CUDA -- NVIDIA GeForce RTX 3050 6GB …) CPU: 12th Gen Intel(R) Core(TM) i5-12450HX	cuda:0 (NVIDIA GeForce RTX 3050 6GB Laptop GPU)	6e9c329f7774…75c88f5c	≠	c849bb0ab6b1…6723a3f2	≠

Bottom-K Selection

PASS — raw ordering diverged, SPCMP converged

Hardware	Device / Algo	Raw SHA-256	=?	SPCMP SHA-256	=?
CPU (CPU -- numpy (GPU bypassed by --hw c…) CPU: Intel(R) Xeon(R) CPU @ 2.20GHz	cpu (numpy) np.argpartition ascending (heap order)	8ecdf89c85fa…9289806b	=	de86a34a316c…b07c5812	=

GPU (CUDA -- NVIDIA GeForce RTX 3050 6GB …) CPU: 12th Gen Intel(R) Core(TM) i5-12450HX	cuda:0 (NVIDIA GeForce RTX 3050 6GB Laptop GPU) torch.topk(largest=False, sorted=True) -> ascending abs order	bf95379ef4e1…767db344	≠	de86a34a316c…b07c5812	=

Appendix — Full SHA-256 Hashes

Top-K Selection

CPU (CPU -- numpy (GPU bypassed by --hw c…) CPU: Intel(R) Xeon(R) CPU @ 2.20GHz [cpu (numpy)]

raw: f76e470f5b7923c3fdb46679d92074847a9fa9f88ac6d630382ffa731ab2d5c4

spcmp: bf476da80bcb9931330471c1564d556db8a2aac3b801ed749e78bf02a9f39551

GPU (CUDA -- NVIDIA GeForce RTX 3050 6GB …) CPU: 12th Gen Intel(R) Core(TM) i5-12450HX [cuda:0 (NVIDIA GeForce RTX 3050 6GB Laptop GPU)]

raw: 3ad9c18dbd0059eabbef6d2fedcf2f72583b5449f61a97113ffb68b1c666ae1a

spcmp: bf476da80bcb9931330471c1564d556db8a2aac3b801ed749e78bf02a9f39551

Scalar Reduction

CPU (CPU -- numpy (GPU bypassed by --hw c…) CPU: Intel(R) Xeon(R) CPU @ 2.20GHz [cpu (numpy)]

raw: 3714dbf099470f16b770a963891807bc49486db508320287eceddedac336978f

spcmp: 66439ad3daef0e3ed8a986b51b6e12670e31acc011b1441239810353668 6665d

GPU (CUDA -- NVIDIA GeForce RTX 3050 6GB …) CPU: 12th Gen Intel(R) Core(TM) i5-12450HX [cuda:0 (NVIDIA GeForce RTX 3050 6GB Laptop GPU)]

raw: 6e9c329f7774bc270184393587d0ca83549567c037f6e7327627bd7a75c88f5c

spcmp: c849bb0ab6b1f0f64d46ca61878ce0b32858f4f487eb9341794e941d6723a3f2

Bottom-K Selection

CPU (CPU -- numpy (GPU bypassed by --hw c…) CPU: Intel(R) Xeon(R) CPU @ 2.20GHz [cpu (numpy)]

raw: 8ecdf89c85fae44cc31c86887f128f47790a9166d507d8747ad4d58a9289806b

spcmp: de86a34a316c5e9a2eb970659e66d83437bf4cba11166e76e11e43e4b07c5812

GPU (CUDA -- NVIDIA GeForce RTX 3050 6GB …) CPU: 12th Gen Intel(R) Core(TM) i5-12450HX [cuda:0 (NVIDIA GeForce RTX 3050 6GB Laptop GPU)]

raw: bf95379ef4e150e329ae7d7d5acfdf2888d5f71da15ac34384ec910d767db344

spcmp: de86a34a316c5e9a2eb970659e66d83437bf4cba11166e76e11e43e4b07c5812

Appendix — Per-Run Details

Run: CPU (20260519T060630Z) — CPU -- numpy (GPU bypassed by --hw cpu)

Top 10,000 elements by absolute value. CPU uses np.argpartition (heap order); GPU uses torch.topk (descending sorted order). Same multiset, different ordering. Raw SHA differs; SPCMP must converge.

Field	Raw	SPCMP
device_used	cpu (numpy)	—
algorithm	np.argpartition (heap order)	—
n_elements	10000	10000
elapsed_s	0.008365	7e-06
sha256	f76e470f5b79…1ab2d5c4	bf476da80bcb…a9f39551

success	—	True

Full reduction of 1,000,000 elements to one scalar. Catastrophic cancellation causes CPU and GPU to produce different VALUES. SPCMP cannot reconcile value-level FP divergence -- documented as the honest limit of the protocol.

Field	Raw	SPCMP
device_used	cpu (numpy)	—
algorithm	—	—
n_elements	1000000	1
elapsed_s	0.000645	0.0
sha256	3714dbf09947…c336978f	66439ad3daef…6686665d
success	—	True

Bottom 10,000 elements by absolute value. CPU uses np.argpartition ascending (heap order); GPU uses torch.topk(largest=False) (ascending sorted order). Same multiset, different ordering. Raw SHA differs; SPCMP must converge.

Field	Raw	SPCMP
device_used	cpu (numpy)	—
algorithm	np.argpartition ascending (heap order)	—
n_elements	10000	10000
elapsed_s	0.011446	7e-06
sha256	8ecdf89c85fa…9289806b	de86a34a316c…b07c5812
success	—	True

Run: GPU (20260519T060652Z) — CUDA -- NVIDIA GeForce RTX 3050 6GB Laptop GPU

Top 10,000 elements by absolute value. CPU uses np.argpartition (heap order); GPU uses torch.topk (descending sorted order). Same multiset, different ordering. Raw SHA differs; SPCMP must converge.

Field	Raw	SPCMP
device_used	cuda:0 (NVIDIA GeForce RTX 3050 6GB Laptop GPU)	—
algorithm	torch.topk(largest=True, sorted=True) -> descending abs order	—
n_elements	10000	10000
elapsed_s	0.125721	4e-06
sha256	3ad9c18dbd00…c666ae1a	bf476da80bcb…a9f39551
success	—	True

Full reduction of 1,000,000 elements to one scalar. Catastrophic cancellation causes CPU and GPU to produce different VALUES. SPCMP cannot reconcile value-level FP divergence -- documented as the honest limit of the protocol.

Field	Raw	SPCMP

device_used	cuda:0 (NVIDIA GeForce RTX 3050 6GB Laptop GPU)	—
algorithm	—	—
n_elements	1000000	1
elapsed_s	0.021971	0.0
sha256	6e9c329f7774…75c88f5c	c849bb0ab6b1…6723a3f2
success	—	True

Bottom 10,000 elements by absolute value. CPU uses np.argpartition ascending (heap order); GPU uses torch.topk(largest=False) (ascending sorted order). Same multiset, different ordering. Raw SHA differs; SPCMP must converge.

Field	Raw	SPCMP
device_used	cuda:0 (NVIDIA GeForce RTX 3050 6GB Laptop GPU)	—
algorithm	torch.topk(largest=False, sorted=True) -> ascending abs order	—
n_elements	10000	10000
elapsed_s	0.000738	4e-06
sha256	bf95379ef4e1…767db344	de86a34a316c…b07c5812
success	—	True

SPCMP v1 — Frozen Semantics. Artifacts are deterministic across CPUs, GPUs, compilers, and driver versions when SPCMP preprocessing is applied prior to PCMP v1 canonical ordering, compression, and hashing.